PHARMACODE READING DIRECTION

Front

110 mm

Back

110 mm

360 mm

Dorzolamide Hydrochloride Eye Drops, Solution
(2% w/v as Dorzolamide)
Monosopt Eye Drops

BRAND OR PRODUCT NAME:
Monosopt Eye Drops

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S):
Dorzolamide Hydrochloride Eye Drops, Solution (2 % w/v as Dorzolamide)

PRODUCT DESCRIPTION:
Clear, colorless to nearly colorless and slightly viscous solution, practically free from particles.

PHARMACODYNAMICS:
Pharmacotherapeutic group: Antiglaucoma preparations and miotic, Carbonic Anhydrase Inhibitors, dorzolamide, ATC code: S01EC03
Carbonic anhydrase (CA) is an enzyme found in many tissues of the body including the eye. It catalyzes the reversible reaction involving the hydration of carbon dioxide and the dehydration of carbonic acid. In humans, carbonic anhydrase exists as a number of isoenzymes, the most active being carbonic anhydrase II (CA-II), found primarily in red blood cells (RBCs), but also in other tissues. Inhibition of carbonic anhydrase in the ciliary processes of the eye decreases aqueous humor secretion, presumably by slowing the formation of bicarbonate ions with subsequent reduction in sodium and fluid transport. The result is a reduction in intraocular pressure (IOP).
Dorzolamide Hydrochloride Ophthalmic Solution contains dorzolamide hydrochloride, an inhibitor of human carbonic anhydrase II. Following topical ocular administration, Dorzolamide Hydrochloride reduces elevated intraocular pressure. Elevated intraocular pressure is a major risk factor in the pathogenesis of optic nerve damage and glaucomatous visual field loss.
Unlike miotics, Dorzolamide eye drop reduces intraocular pressure without the common side effects of miotics such as night blindness, accommodative spasm and pupillary constriction. Unlike topical beta-blockers, Dorzolamide eye drop has minimal or no effect on pulse rate or blood pressure. Topically applied beta-adrenergic blocking agents also reduce IOP by decreasing aqueous humor secretion but by a different mechanism of action. Studies have shown that when Dorzolamide eye drop is added to a topical beta-blocker, additional reduction in IOP is observed; this finding is consistent with the reported additive effects of beta-blockers and oral carbonic anhydrase inhibitors.

PHARMACOKINETICS:
Absorption
When topically applied, dorzolamide reaches the systemic circulation. To assess the potential for systemic carbonic anhydrase inhibition following topical administration, drug and metabolite concentrations in RBCs and plasma and carbonic anhydrase inhibition in RBCs were measured.
Distribution
Dorzolamide accumulates in RBCs during chronic dosing as a result of binding to CA-II. The parent drug forms a single N-desethyl metabolite, which inhibits CA-II less potently than the parent drug but also inhibits CA-I. The metabolite also accumulates in RBCs where it binds primarily to CA-I. Plasma concentrations of dorzolamide and metabolite are generally below the assay limit of quantitation (15nM). Dorzolamide binds moderately to plasma proteins (approximately 33%).
Elimination
Dorzolamide is primarily excreted unchanged in the urine; the metabolite also is excreted in urine. After dosing is stopped, dorzolamide washes out of RBCs nonlinearly, resulting in a rapid decline of drug concentration initially, followed by a slower elimination phase with a half-life of about four months.

INDICATIONS:
Dorzolamide Hydrochloride Ophthalmic Solution is indicated in the treatment of elevated intraocular pressure in patients with ocular hypertension and open-angle glaucoma.

RECOMMENDED DOSAGE:
When used as monotherapy, the dose is 1 drop of Dorzolamide eye drops in the affected eye(s) 3 times daily. When used as adjunctive therapy with an ophthalmic β-blocker, the dose is 1 drop of Dorzolamide eye drops in the affected eye(s) 2 times daily. When substituting Dorzolamide eye drops for another ophthalmic anti-glaucoma agent, discontinue the other agent after proper dosing on 1 day, and start Dorzolamide eye drops on the next day. If more than one topical ophthalmic drug is being used, the drugs should be administered at least 10 min apart.

CONTRAINDICATIONS:
Dorzolamide Hydrochloride is contraindicated in patients who are hypersensitive to any component of this product Dorzolamide has not been studied in patients with severe renal impairment (CrCl < 30 ml/min) or with hyperchloraemic acidosis. Because dorzolamide and its metabolites are excreted predominantly by the kidney, dorzolamide is therefore contraindicated in such patients.

WARNINGS AND PRECAUTIONS:
1) If signs of serious reactions (ie. Stevens-Johnson syndrome and toxic epidermal and toxic epidermal necrolysis) or hypersensitivity occur, discontinue the use of Dorzolamide eye drop solution. Local ocular adverse effects, primarily conjunctivitis and lid reactions, were reported with chronic administration of Dorzolamide eye drop solution. Some of these reactions had the clinical appearance and course of an allergic-type reaction that resolved upon discontinuation of drug therapy. If such reactions are observed, discontinuation of treatment with Dorzolamide eye drop solution should be considered. There is a potential for an additive effect on the known systemic effects of carbonic anhydrase inhibition in patients receiving an oral carbonic anhydrase inhibitor and Dorzolamide eye drop solution. The concomitant administration of Dorzolamide eye drop and oral carbonic anhydrase inhibitors has not been studied and is not recommended. Choroidal detachment has been reported with administration of aqueous suppressant therapy (e.g., dorzolamide) after filtration procedures.
2) Therapy with oral carbonic anhydrase inhibitors has been associated with urolithiasis as a result of acid-base disturbances, especially in patients with a prior history of renal calculi. Although no acid-base disturbances have been observed with dorzolamide, urolithiasis has been reported infrequently. Because dorzolamide is a topical carbonic anhydrase inhibitor that is absorbed systemically, patients with a prior history of renal calculi may be at increased risk of urolithiasis while using dorzolamide.
3) Paediatric patients: Dorzolamide has not been studied in patients less than 36 weeks gestational age and less than one week of age. Patients with significant renal tubular immaturity should only receive dorzolamide after careful consideration of the risk benefit balance because of the possible risk of metabolic acidosis.
4) Dorzolamide eye drop solution contains the preservative benzalkonium chloride, which may be absorbed by soft contact lenses. Therefore, Dorzolamide eye drop solution should not be administered while wearing soft contact lenses. The contact lenses should be removed before application of the drops and not be reinserted earlier than 15 min after use.

Effects on Ability to Drive and Use Machine
No studies on the effects on the ability to drive and use machines have been performed.
Possible side effects such as dizziness and visual disturbances may affect the ability to drive and use machines.

INTERACTIONS WITH OTHER MEDICAMENTS:
Specific drug interaction studies have not been performed with dorzolamide eye drops. Dorzolamide eye drop was used concomitantly with the following medications without evidence of adverse interactions: Timolol ophthalmic solution, betaxolol ophthalmic solution and systemic medications, including ACE-inhibitors, calcium-channel blockers, diuretics, nonsteroidal anti-inflammatory drugs including aspirin, and hormones (e.g., estrogen, insulin, thyroxine).

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION
Pregnancy
There are no adequate and well-controlled studies in pregnant women. Dorzolamide Hydrochloride should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.
Lactation
It is not known whether this drug is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from Dorzolamide Hydrochloride, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

ADVERSE EFFECTS/ UNDESIRABLE EFFECTS:
Dorzolamide Hydrochloride was evaluated in more than 1400 individuals in controlled and uncontrolled clinical studies. In long term studies of 1108 patients treated with Dorzolamide Hydrochloride as monotherapy or as adjunctive therapy with an ophthalmic beta-blocker, the most frequent cause of discontinuation (approximately 3%) from treatment with Dorzolamide Hydrochloride was drug-related ocular adverse reactions, primarily conjunctivitis and lid reactions.
The following adverse reactions have been reported either during clinical trials or during post-marketing experience with dorzolamide:
Nervous system disorders:
Common: headache
Rare: dizziness, paraesthesia
Eye disorders:
Very common: burning and stinging
Common: superficial punctate keratitis, tearing, conjunctivitis, eyelid inflammation, eye itching, eyelid irritation, blurred vision
Uncommon: iridocyclitis
Rare: irritation including redness, pain, eyelid crusting, transient myopia (which resolved upon discontinuation of therapy), corneal oedema, ocular hypotony, choroidal detachment following filtration surgery
Not known: foreign body sensation in eye
Cardiac disorders:
Not known: palpitations
Respiratory, thoracic, and mediastinal disorders:
Rare: epistaxis
Not known: dyspnoea
Gastrointestinal disorders:
Common: nausea, bitter taste
Rare: throat irritation, dry mouth
Skin and subcutaneous tissue disorders:
Rare: contact dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis
Renal and urinary disorders:
Rare: urolithiasis
General disorders and administration site conditions:
Common: asthenia/fatigue
Rare: hypersensitivity: signs and symptoms of local reactions (palpebral reactions) and systemic allergic reactions, including angioedema, urticaria and pruritus, rash, shortness of breath, rarely bronchospasm
Investigations:
Dorzolamide was not associated with clinically meaningful electrolyte disturbances.

OVERDOSAGE AND TREATMENT:
Treatment should be symptomatic and supportive. Electrolyte imbalance, development of an acidotic state, and possible central nervous system effects may occur. Serum electrolyte levels (particularly potassium) and blood pH levels should be monitored.

STORAGE CONDITIONS:
Store below 30°C. Protect from light.

SHELF LIFE:
24 months from the date of manufacture. This medicine should be used within 28 days of opening the bottle. Discard any remaining solution four weeks after opening

DOSAGE FORMS AND PACKAGING AVAILABLE:
Eye drop, Solution
Pack Style:- 5 ml in 5 ml Three Piece bottle (1 x 5ml Quantity)
Packing configuration: - 5 ml in 5 ml sterile white opaque LDPE container with white opaque open sterile nozzle & HDPE orange colour TSTR Tear off cap, one such container packed in carton with leaflet.

Reg. No. : MAL24046026AZ

NAME AND ADDRESS OF MANUFACTURER/ PRODUCT REGISTRATION HOLDER:
MICRO LABS LIMITED
Plot Nos. 113 to 116, IV Phase, KIADB,
Bommasandra Industrial Area,
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PRODUCT REGISTRATION HOLDER
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Kuala Lumpur, Malaysia

DATE OF REVISION OF PI:
Sep. 2024

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Size: 110 (L) x 360 (W) mm
(Folded size: 110 mm (L) x 23±1 mm (W) x NMT 1.25 mm (Thickness))
Pharma Print Paper 35-50 gsm

Font: Times New Roman
Font size: 7

PHARMACODE READING DIRECTION

14 (L) x 6 (H) mm

PHARMACODE READING DIRECTION

2568